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Here are summaries of the core problem-solving methods from Data 140, explained with examples.

1

Modeling (Construction/Deconstruction) of Random Variable Distribution

Modeling is understanding. Modeling of random distribution is to tell what a distribution is in terms of the available understandings. In this regard, modeling may also be called expression, expressing concept A in terms of other concepts that have been understood, or better understood.

One of the often powerful expressions is linear combination.

1.1 Randomness Characterizations

(We can't know everything of a random variable's distribution, then we try to get some partial understanding of random distributions, as much as possible.)

Mean the weighted average of all possible values with the weights being the possibilities of the corresponding values. Showing the "center" of the value.

$$E[X - E[X]] = 0$$

Variance Variance measures the "spread" of a random variable around its mean ("center"). Mathematically, it is the expected squared deviation from the mean: (Why squared? indifferent to sign, want it to be non-negative) (Why not use other operation to get non-negative? e.g. absolute value $|X - E[X]|$? We want the transformation/operation to be continuous, to be possible derivation, for example, for finding the maximum/minimum of the variance) (mathematically interesting: the difference between the mean of the square of X and the square of the mean of X)

$$Var(X) = E[(X - E[X])^2]$$

$$Var(X) = E(X^2) - (E[X])^2$$

Simplification

Standard deviation the square root of the variance (Why perform square root operation on the variance? to be more comparable to the random variable's values, and its mean.)

$$SD = \sqrt{Var(X)}$$

Co-Variance

MGF

1.2 Building blocks

1.2.1 Bernoulli distribution (p probability of success)

$P(X = k), k \in \{1, 0\}, P(X = 1) = p$, then $P(X = 0) = 1 - p$, called q

- The simplest random distribution:
 - Only two outcomes (as one outcome wouldn't be random!)
- Express, relate between the outcome of 1 and 0, p to q .
- Mean: $E(X)$: the average of all possible values. Only two possibilities, 1, 0, then the average of them: $1 \times p + 0 \times (1 - p)$ this evaluates to p

Bernoulli $E[X] = p$

- Variance: $Var(X) = E(X^2) - (E[X])^2 = E(X^2) - p^2$ by simplified calculation

Need to figure out $E(X^2)$ (we know $E(X)$ does not automatically implies that we know $E(X^2)$, it definitely $E(X^2) \neq E(X)^2$)

When there is no better means. Try the brutal force of calculating from the definition:

$E(X^2) = \sum_{\text{all possible values of } X^2} p_{X^2} X^2$ where p_{X^2} is the probability of a new random variable Y , $P(Y = X^2)$ as X^2 is a (1-to-1) function, thus $p_{X^2} = P(Y = k^2) = P(X = k) = p$ therefore, $E(X^2) = p1^2 + (1 - p)0^2 = p$ therefore, $Var(X) = E(X^2) - p^2 = p - p^2 = p(1 - p) = pq$

Bernoulli: $E(X^2) = p$

Bernoulli: $Var(X) = pq$

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1.2.2 Hypergeometric distribution

The Hypergeometric distribution models the number of successes (g) in a fixed number of draws (n) from a finite population (N) when sampling is done without replacement.

Key Characteristics Nature of Trials: Unlike the Binomial distribution, the trials are not independent because the probability of success changes with every draw as items are removed from the population. Parameters: It is defined by the population size (N), the total number of successes available in that population (G), and the number of items sampled (n). Probability Mass Function (PMF):

$$P(X = g) = \frac{\binom{G}{g} \binom{N-G}{n-g}}{\binom{N}{n}}$$

where g is the number of successes in your sample.

Statistical Properties Mean: $n \cdot \frac{G}{N}$. Variance: $n \cdot \frac{G}{N} \cdot \frac{N-G}{N} \cdot \frac{N-n}{N-1}$. **Finite Population Correction:** The factor $\frac{N-n}{N-1}$ in the variance formula accounts for the reduction in variability caused by sampling a large portion of a finite population without replacement.

1. The derivation of hypergeometric

The derivation of the Hypergeometric PMF is based on counting combinations when sampling without replacement from a finite population.

- (a) 1. The Scenario Suppose you have a population of size N that contains G “good” items (successes) and B “bad” items (failures), where $N = G + B$. You take a simple random sample of size n without replacement.

The random variable X represents the number of successes in your sample.

- (b) 2. The Combinatorial Logic To find $P(X = g)$, we use the “favorable outcomes over total outcomes” approach:

Total ways to choose the sample There are $\binom{N}{n}$ ways to choose any n items from the total population of N .

Ways to choose exactly g successes There are $\binom{G}{g}$ ways to pick g successes from the G available in the population.

Ways to choose the remaining failures Since you picked g successes, you must pick $n - g$ failures from the B available failures. There are $\binom{B}{n-g}$ ways to do this.

- (c) 3. The Mass Function Multiplying the ways to get successes and failures gives the total favorable outcomes. Dividing by the total possible samples yields the PMF:

$$P(X = g) = \frac{\binom{G}{g} \binom{B}{n-g}}{\binom{N}{n}}$$

This formula is valid for $0 \leq g \leq n$, provided $g \leq G$ and $n - g \leq B$.

Does this combinatorial approach make sense, or would you like to see how this distribution’s **expected value** ($n \cdot \frac{G}{N}$) is derived?

2. consecutive odds ratio of hypergeometric

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3. the difference between binomial and hypergeometric

The main difference between the Binomial and Hypergeometric distributions lies in the independence of trials and the sampling method:

Binomial Distribution Models the number of successes in n independent trials where sampling is done with replacement.

- Because items are replaced, the probability of success remains constant for every trial.

Hypergeometric Distribution Models the number of successes in n dependent trials where sampling is done without replacement from a finite population.

- The probability of success changes with each draw based on previous outcomes (not constant probability).

Comparison of Properties

Property	Binomial (n, p)	Hypergeometric (N, G, n)
Independence	Independent trials	Dependent trials
Probability	Constant (p)	Changes with each draw
Mean	np	$n \cdot \frac{G}{N}$
Variance	$np(1 - p)$	includes a finite population correction factor: $\frac{N-n}{N-1}$

When sampling a small fraction from a very large population, the dependence in the Hypergeometric distribution becomes negligible, and the Binomial distribution serves as a close approximation.

Does this help clarify why we use the “without replacement” distinction? Would you like to see how the Hypergeometric consecutive odds ratio is derived next? entered on: 2026-05-04 08:10:18

1.3 Using better known distribution as building blocks to study other distributions

The most classic example in Data 140 is the Binomial distribution, which is constructed entirely from Bernoulli trials.

1.3.1 Building the Binomial Distribution

A Binomial (n, p), n Bernoulli (p) trials. (Repeated Bernoulli experiments)

The Component: Each trial X_i is a Bernoulli random variable where $X_i = 1$ (success) or $X_i = 0$ (failure). The Construction: By adding them together

you create a new distribution that counts the total number of successes in those n trials. Why this works: This “building block” approach allows you to derive the properties of the Binomial distribution directly from the simpler Bernoulli trial. For instance, because expectation is linear, the mean of the Binomial is simply $n \times E[X] = np$.

The Probability Mass (Density) Function (PMF) of binomial: – To derive the **Probability Mass Function (PMF)** for the Binomial(n, p) distribution, we view it as a sequence of n independent Bernoulli trials.

$$\text{Binomial: } S_n = \sum_{i=1}^n X_i$$

X_i Bernoulli i.i.d.

$$\text{Binomial } E[X] = np$$

1. **1. Probability of a Single Specific Sequence** Imagine a specific sequence of n trials that results in exactly k successes and $n - k$ failures (e.g., $S, S, F, \dots$). Because the trials are independent, we multiply their individual probabilities: The k successes each have probability p , resulting in p^k . The $n - k$ failures each have probability $q = (1 - p)$, resulting in $(1 - p)^{n-k}$. The probability of that one specific sequence is $p^k(1 - p)^{n-k}$.
2. **2. Counting All Possible Sequences** There are many different sequences that can result in k successes. The number of ways to choose exactly k positions for successes out of n available trials is given by the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

3. **3. The Final Density (PMF)** Since each of these $\binom{n}{k}$ sequences is mutually exclusive and has the same probability, we sum them to get the final formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \text{ for } k = 0, 1, \dots, n$$

1.3.2 Building Geometric Distribution

You can also use Bernoulli trials to define “waiting time” distributions: The number of Bernoulli trials needed to get the first success. $P(X = k) = p(1 - p)^{(k-1)}$ where p is the probability of success.

1.3.3 Building Negative Binomial Distribution

The number of Bernoulli trials needed to reach a fixed number of successes.

To derive the Probability Mass Function (PMF) for the negative binomial distribution, we define the random variable X as the total number of trials needed to achieve a fixed number of successes, r .

1. **1. The Logical Requirement** For the r -th success to occur exactly on total trial k , two conditions must be met:
 - (a) In the first $k - 1$ trials, there must be exactly $r - 1$ successes.
 - (b) The k -th trial must be a success.
2. **2. Probability of the First $k - 1$ Trials** So the first part follows a Binomial distribution. We need to choose $r - 1$ positions for successes out of $k - 1$ available trials: Ways to arrange: $\binom{k-1}{r-1}$ Probability of $r - 1$ successes: p^{r-1} Probability of $(k - 1) - (r - 1) = k - r$ failures: $(1 - p)^{k-r}$.
This gives us: $\binom{k-1}{r-1} p^{r-1} (1 - p)^{k-r}$.

Binomial: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
for $k = 0, 1, \dots, n$
 $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Geometric distribution (p): X
 $P(X = k) = p(1 - p)^{(k-1)}$

3. 3. The Final Trial The k -th trial must be a success, which has probability p . Since the trials are independent, we multiply this by our previous result:

$$P(X = k) = \left[\binom{k-1}{r-1} p^{r-1} (1-p)^{k-r} \right] \times p$$

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r} \text{ for } k = r, r+1, \dots$$

4. Alternative Parameterization In some contexts, the distribution is defined by the number of failures (k) observed before the r -th success. In that case, the total trials are $r + k$, and the PMF is written as:

$$P(X = k) = \binom{n+k-1}{k} p^r (1-p)^k$$

5. Why it is Negative Binomial? The Negative Binomial distribution is named that way primarily because it answers the “inverse” or “negative” version of the question posed by the Binomial distribution.

Here are the two main reasons for the name:

- (a) Inverse Random Variables: In a Binomial distribution, you fix the number of trials and count the random number of successes.

In a Negative Binomial distribution, you do the opposite: you fix the number of successes you want to achieve and count the random number of trials it takes to get there.

Negative Binomial: X

$$P(X = k) =$$

$$\binom{k-1}{r-1} p^r (1-p)^{k-r}$$

for $k = r, r+1, \dots$

k : the number of total trials

Two formulations of negative binomial (geometric):

1. (k : total number of trials to achieve fixed number r success)

2. (k : number of failures to achieve fixed number r success)

geometric: is negative binomial with $r = 1$ constant omitted

Distribution	Fixed Quantity	Random Variable	Interpretation	Relationships
Binomial	# of trials (n)	# of successes (k)	Count successes in a fixed # of trials	$n \geq k$
Negative Binomial	# of successes (r)	# of trials (k)	Count trials needed to reach a fixed # of successes	$k \geq r$
Negative Binomial	# of successes (r)	or # of failures (k)	Count failures needed to reach a fixed # of successes	$k + r = \text{total trials}$

Both distributions assume independent Bernoulli trials with constant p

Practical modeling insight

Use Binomial when your experiment budget is fixed. Use Negative Binomial when your stopping condition is success-based (common in reliability testing, queuing, and retry systems).

If you want, I can map this to likelihood functions and show how they differ in inference workflows.

1.3.4 Negative Binomial degenerates (collapses) to Geometric distribution

To show that the Geometric distribution is a special case of the Negative Binomial distribution, we simply set the required number of successes (r) to 1 in the Probability Mass Function (PMF).

1. The Mathematical Substitution

(a) Negative Binomial PMF: This distribution models the number of trials (k) to get r successes:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

(b) Set $r = 1$:

$$P(X = k) = \binom{k-1}{1-1} p^1 (1-p)^{k-1}$$

(c) Simplify:

- $\binom{k-1}{0} = 1$ (There is only one way to have the first success on the k -th trial: $k-1$ failures followed by one success).
- Result: $P(X = k) = p(1-p)^{k-1}$.

This final expression is exactly the PMF for the Geometric distribution, defined as the number of trials until the first success.

2. Conceptual Link

- Negative Binomial: “How many trials until I get r successes?”.
- Geometric: “How many trials until I get the first success?”.

When $r = 1$, the questions are identical.

1.3.5 Law of Small Numbers: Poisson as approximation of the Binomial distribution

“rare events” where the number of trials (n) is large and the probability of success (p) is very small, such that np converges to a constant λ .

Detail development: The **Law of Small Numbers** describes how the Binomial distribution converges to the Poisson distribution when the number of trials n is very large and the probability of success p is very small, such that their product $np = \lambda$ remains constant.

1. **Mathematical Derivation** To derive this, we start with a variable $X_n \sim \text{Binomial}(n, \lambda/n)$ and take the limit as $n \rightarrow \infty$:

Rare event of Binomial X :
 $\lim_{n \rightarrow \infty, p \rightarrow 0} np = \lambda$
 for Binomial (n, p) X $E[X] = np$
 for Binomial X_n
 $\lim_{n \rightarrow \infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

(a) Set up the Binomial PMF:

$$P(X_n = k) = \frac{n(n-1) \dots (n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

(b) Rearrange the terms:

$$P(X_n = k) = \frac{\lambda^k}{k!} \left[\frac{n}{n} \cdot \frac{n-1}{n} \dots \frac{n-k+1}{n} \right] \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$

(c) Evaluate the limits as $n \rightarrow \infty$:

- The product in the brackets $\left[\frac{n}{n} \cdot \frac{n-1}{n} \dots \frac{n-k+1}{n}\right]$ consists of k terms that each approach 1.
- The term $\left(1 - \frac{\lambda}{n}\right)^n$ approaches $e^{-\lambda}$.
- The term $\left(1 - \frac{\lambda}{n}\right)^{-k}$ approaches 1 because k is a fixed constant while n grows.

(d) Final Result: By combining these limits, we arrive at the Poisson PMF:

$$\lim_{n \rightarrow \infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

This approximation is considered accurate in practice when n is large and the average λ is relatively small.

2. Poisson Distribution Definition A random variable X follows a Poisson (λ) distribution if its Probability Mass Function (PMF) is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \text{ for } k = 0, 1, 2, \dots$$

where $\lambda > 0$ is the average rate of occurrence.

The Poisson distribution was developed as a mathematical limit of the **Binomial distribution**, discovered by Siméon Denis Poisson in 1837.

(a) **The Core Intuition: “The Law of Small Numbers”** The primary intuition source is the study of **rare events** occurring within a large number of independent trials. **The Limit:** If you have a sequence of trials (n) that goes to infinity while the probability of success (p) goes to zero, but their product ($np = \lambda$) stays constant, the distribution converges to Poisson. **The “Building Block”:** It essentially treats a continuous region of time or space as being divided into many tiny “bins” where a “success” (an event) is extremely unlikely in any single bin.

(b) **Historical Real-World Sources** The distribution became famous through observations of phenomena that fit this “rare event” model perfectly: **Radioactivity:** Counting the clicks on a Geiger counter over time. **Military Data:** The number of Prussian soldiers killed by horse kicks annually across different army corps. **Wartime Clustering:** The hits from Nazi V1 “Buzz Bombs” in London during WWII, where the city was divided into small sectors to see if the bombs were clustering or falling randomly.

Poisson X : $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$
for $k = 0, 1, 2, \dots$
 $\lambda > 0$ the average rate of occurrence

Poisson: rare events of arrival to some “bins” over (long) time or space with average rate of λ

(c) **Key Logical Assumptions** For this intuition to hold, four things must be true:

- i. **Independence:** One event does not affect the likelihood of another.
- ii. **Constant Rate:** Events occur at a steady average frequency (λ).
- iii. **Proportionality:** The number of events is proportional to the size of the interval.
- iv. **Non-Simultaneity:** Two events cannot happen at the exact same instant or location.

1.3.6 Logarithmic Series distribution (not required in Data 140)

The Logarithmic Series distribution is a discrete probability distribution used to model counts where the value zero is not possible, meaning its support is $k = 1, 2, 3, \dots$

1. Probability Mass Function (PMF) As a member of the **power series family**, its PMF is defined by the parameter θ ($0 < \theta < 1$):

$$P(X = k) = \frac{\theta^k}{-k \ln(1 - \theta)}$$

In this structure, the “coefficient” is $a(k) = 1/k$ and the “series function” is $f(\theta) = -\ln(1 - \theta)$.

2. Meaning and Intuition

- **Species Abundance:** It is most famous for modeling biological diversity, such as the number of individual insects found for each species in a collection.
- **Zero-Truncated Limit:** Intuitively, you can think of it as the limiting form of a Negative Binomial distribution that has been “zero-truncated” (the $k = 0$ case is removed).
- **Long-Tailed Behavior:** It is a “long-tailed” distribution, meaning it can model scenarios where most categories have very few members, but a few rare categories have very high counts.

3. Visual Diagnosis In an Ord Plot, the Logarithmic Series is identified by a positive slope ($b > 0$) and a negative intercept ($a < 0$).

1.3.7 Power series family

The **power series family** is a broad group of discrete probability distributions where the probability of a value k is defined by the formula $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$.

In this structure, θ is a positive parameter, $a(k)$ is a coefficient function of k , and $f(\theta)$ is the “series function” that ensures the probabilities sum to one.

Power series family:

$$P(X = k) = \frac{a(k)\theta^k}{f(\theta)}, \theta > 0$$

1. Log-Ratio of Power Series Distributions Given $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$, then $R(k) = \frac{P(k)}{P(k-1)} = \frac{a(k)}{a(k-1)}\theta$ significantly simplified!

ratio of consecutive probabilities:

$$\frac{P(k)}{P(k-1)} = \frac{a(k)}{a(k-1)}\theta$$

2. Components of Power Series Distributions

Distribution	$P(X = k)$	Coefficient ($a(k)$)	Parameter (θ)	Series Function ($f(\theta)$)
Poisson (k : # of successes out of infinity)	$\frac{e^{-\lambda}\lambda^k}{k!}$	$1/k!$	λ	e^θ
Binomial (k : # of successes of n trials)	$\binom{n}{k}p^k(1-p)^{n-k}$	$\binom{n}{k}$	$p/(1-p)$	$(1+\theta)^n$
Negative Binomial (k : total # of trials to achieve fixed # r success)	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$	$\binom{k-1}{r-1}$	$1-p$	$\left(\frac{\theta}{(1-\theta)}\right)^r$
Negative Binomial (k : # of failures to achieve fixed # r success)	$\binom{r+k-1}{r-1}p^r(1-p)^k$	$\binom{r+k-1}{r-1}$	$1-p$	$(1-\theta)^{-r}$
Geometric (k : # of total trials till success)	$p(1-p)^{(k-1)}$	1	$1-p$	$\frac{\theta}{(1-\theta)}$
Geometric (k : # of failures till success)	$p(1-p)^{(k)}$	1	$1-p$	$(1-\theta)^{-1}$
Logarithmic Series	$\frac{\theta^k}{-k \ln(1-\theta)}$	$1/k$	θ	$-\ln(1-\theta)$

More components of the power series distributions:

Distribution	$R(k) = \frac{P(k)}{P(k-1)} = \frac{a(k)}{a(k-1)}\theta$	Linear Recurrence $kR(k) = ak + b$
Poisson (k : # of successes out of infinity)	$\frac{\theta}{k} = \frac{\lambda}{k}$	$\lambda = 0k + \lambda$
Binomial (k : # of successes of n trials)	$R(k) = \frac{n-k+1}{k} \cdot \frac{p}{1-p}$	$\left(\frac{-p}{1-p}\right)k + \frac{p(n+1)}{1-p}$
Negative Binomial (k : total # of trials to achieve fixed # r success)	$\frac{k-1}{k-r}(1-p)$ or let $k' = k - r$, $R(k') = \frac{k'+r-1}{k'}(1-p)$ In fact, k' is just the # of failures!	(hard to convert to $ak + b$ form, use the equivalent with k as # of failures)
Negative Binomial (k : # of failures to achieve fixed # r success)	$R(k) = \frac{k+r-1}{k} \cdot (1-p)$	$(1-p)k' + (r-1)(1-p)$
Geometric (k : # of total trials till success)	$\theta = 1-p$	$(1-p)k + (r-1)(1-p)$
Geometric (k : # of failures till success)	$\theta = 1-p$	$(1-p)k$
Logarithmic Series	$\frac{k-1}{k}\theta$	$(1-p)k$ $\theta k - \theta$

3. Identifying the components of a power series distribution $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$:

(a) Overall receipt:

- Identify $a(k)$: Isolate the coefficient function that depends on k but is not part of the term with k as an exponent.
- Identify θ^k : Group all terms where k is the power; the base of this expression is your parameter θ .

iii. Determine $f(\theta)$: The remaining terms (which do not depend on k) represent $\frac{1}{f(\theta)}$. Take their reciprocal to find the series function $f(\theta)$.

(b) Poisson In $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$:

- i. $a(k) = 1/k!$ leaving $e^{-\lambda} \lambda^k$
- ii. $\theta = \lambda$ leaving $e^{-\lambda}$
- iii. $1/f(\theta) = e^{-\lambda} \Rightarrow f(\theta) = e^\theta$.

(c) Binomial

In $\binom{n}{k} p^k (1-p)^{n-k}$

- i. $a(k) = \binom{n}{k}$ leaving $p^k (1-p)^{n-k}$ rearranging $p^k (1-p)^{n-k} = \left[\frac{p}{1-p} \right]^k (1-p)^n$
- ii. $\theta = \frac{p}{1-p}$ leaving $(1-p)^n$
- iii. $1/f(\theta) = (1-p)^n$ needing to express $1-p$ in terms of $\theta = \frac{p}{1-p}$

Trying to construct relationship between $1-p$ and θ .

We have $p + (1-p) = 1$ creating the appearance of θ in the equation, by deviding $1-p$ at the both side of the above equation, resulting another equation:

$$\frac{p}{1-p} + 1 = \frac{1}{1-p}$$

thus we have $\theta + 1 = \frac{1}{1-p}$ therefore, we have $1-p = \frac{1}{\theta+1}$

thus

$$1/f(\theta) = (1-p)^n = \frac{1}{(\theta+1)^n}$$

Conclusion: $f(\theta) = (\theta+1)^n$

(d) Negative binomial

In $\binom{k-1}{r-1} p^r (1-p)^{k-r}$ (k : total # of trials to achieve fixed # r success)

- i. $a(k) = \binom{k-1}{r-1}$ leaving $p^r (1-p)^{k-r}$ rearranging $p^r (1-p)^{k-r} = (1-p)^k p^r (1-p)^{-r} = (1-p)^k \left(\frac{p}{1-p} \right)^r$
- ii. $\theta = 1-p$ leaving $\left(\frac{p}{1-p} \right)^r = \left(\frac{1-\theta}{\theta} \right)^r$
- iii. $1/f(\theta) = \left(\frac{1-\theta}{\theta} \right)^r$ therefore $f(\theta) = \left(\frac{\theta}{1-\theta} \right)^r$

In $\binom{r+k-1}{r-1} p^r (1-p)^k$ (k : # of failures to achieve fixed # r success)

- i. $a(k) = \binom{r+k-1}{r-1}$ leaving $p^r (1-p)^k$

tricks: with $\theta = \frac{p}{1-p}$
 use $p + (1-p) = 1$
 to get $\frac{p}{1-p} + 1 = \frac{1}{1-p}$
 $\theta + 1 = \frac{1}{1-p}$

- ii. $\theta = 1 - p$ leaving p^r as $p = 1 - \theta$
- iii. $1/f(\theta) = (1 - \theta)^r$ therefore $f(\theta) = (1 - \theta)^{-r}$

(e) Geometric

For k to be the number of trials to get a success $k = 1, 2, \dots$ In $p(1 - p)^{(k-1)}$

- i. $a(k) = 1$ remaining $p(1 - p)^{(k-1)}$
- ii. $\theta = 1 - p$ remaining $p(1 - p)^{-1}$
- iii. $\frac{1}{f(\theta)} = p(1 - p)^{-1}$ expressing $p = 1 - \theta$, then $p(1 - p)^{-1} = \frac{(1-\theta)}{\theta}$ therefore, $f(\theta) = \frac{\theta}{(1-\theta)}$

For k to be the number of failures before a success $k = 0, 1, 2, \dots$ In $p(1 - p)^{(k)}$

- i. $a(k) = 1$ remaining $p(1 - p)^{(k)}$
- ii. $\theta = 1 - p$ remaining p
- iii. $\frac{1}{f(\theta)} = p$ expressing $p = 1 - \theta$, then therefore, $f(\theta) = (1 - \theta)^{-1}$

(f) Logarithmic Series

$$\ln \frac{\theta^k}{-k \ln(1-\theta)}$$

- i. $a(k) = \frac{1}{k}$ leaving $\frac{\theta^k}{-\ln(1-\theta)}$
- ii. $\theta = \theta$ leaving $\frac{1}{-\ln(1-\theta)}$
- iii. $\frac{1}{f(\theta)} = \frac{1}{-\ln(1-\theta)}$ therefore, $f(\theta) = -\ln(1 - \theta)$

4. Application of Power Series Distributions

- (a) Mode of a distribution (distribution value of peak probability) The mode of a discrete distribution is a possible value of the distribution that has the highest probability. In a continuous distribution, it is the value where the probability density is at its peak. Key points to remember: Multiple Modes: A distribution can have more than one mode if multiple values share the highest probability. Binomial Mode: For a Binomial (n, p) distribution, the mode is the integer part of $(n + 1)p$. Identification: In Data 140, modes are often identified using consecutive odds ratios; as long as the ratio $R(k) \geq 1$, the distribution is rising or flat, and the point where it begins to fall identifies the mode.
- mode of a distribution: distribution value(s) of peak probability
- i. odds ratio (for finding mode) In the context of probability distributions like those studied in Data 140, an odds ratio (specifically a consecutive odds ratio) is the ratio of the probability of a value k to the probability of the preceding value $k - 1$.
- ### Mathematical Definition For $k \geq 1$, the k -th consecutive odds ratio is defined as:

$$R(k) = \frac{P(k)}{P(k-1)}$$

Why It Is Useful Simplified Computation: It allows you to calculate the entire distribution recursively starting from $P(0)$ without having to compute large factorials or powers for every value,. Determining Shape: Comparing $R(k)$ to 1 identifies the “direction” of the distribution’s histogram,:

- $R(k) > 1$: The histogram is rising ($P(k) > P(k-1)$).
- $R(k) = 1$: The histogram is flat.
- $R(k) < 1$: The histogram is falling ($P(k) < P(k-1)$).

Finding the Mode: The largest k for which $R(k) \geq 1$ identifies the mode (the most probable value) of the distribution,.

In categorical data analysis, the term is also used as a measure of association in 2×2 tables, comparing the odds of an event occurring in one group versus another.

Does this clarify how we use these ratios to “build” a distribution, or would you like to see the specific formula for the **Hypergeometric** odds ratio? entered on: 2026-05-04 07:58:23

(b) Usefulness of power series (abstraction) with its log-ratios The power series relationships are particularly useful because they allow us to linearize the probability mass function for graphical diagnosis.

i. 1. Odds Ratios vs. Log-Ratios

Consecutive Odds Ratio the raw ratio $R(k) = \frac{P(k)}{P(k-1)}$. It is used primarily for recursive calculation

- building the distribution step-by-step and
- finding the mode (where the ratio crosses 1).

Log-Ratios the logarithm of that ratio: $\log(R(k))$. Taking the log is the specific “trick” used for linearization.

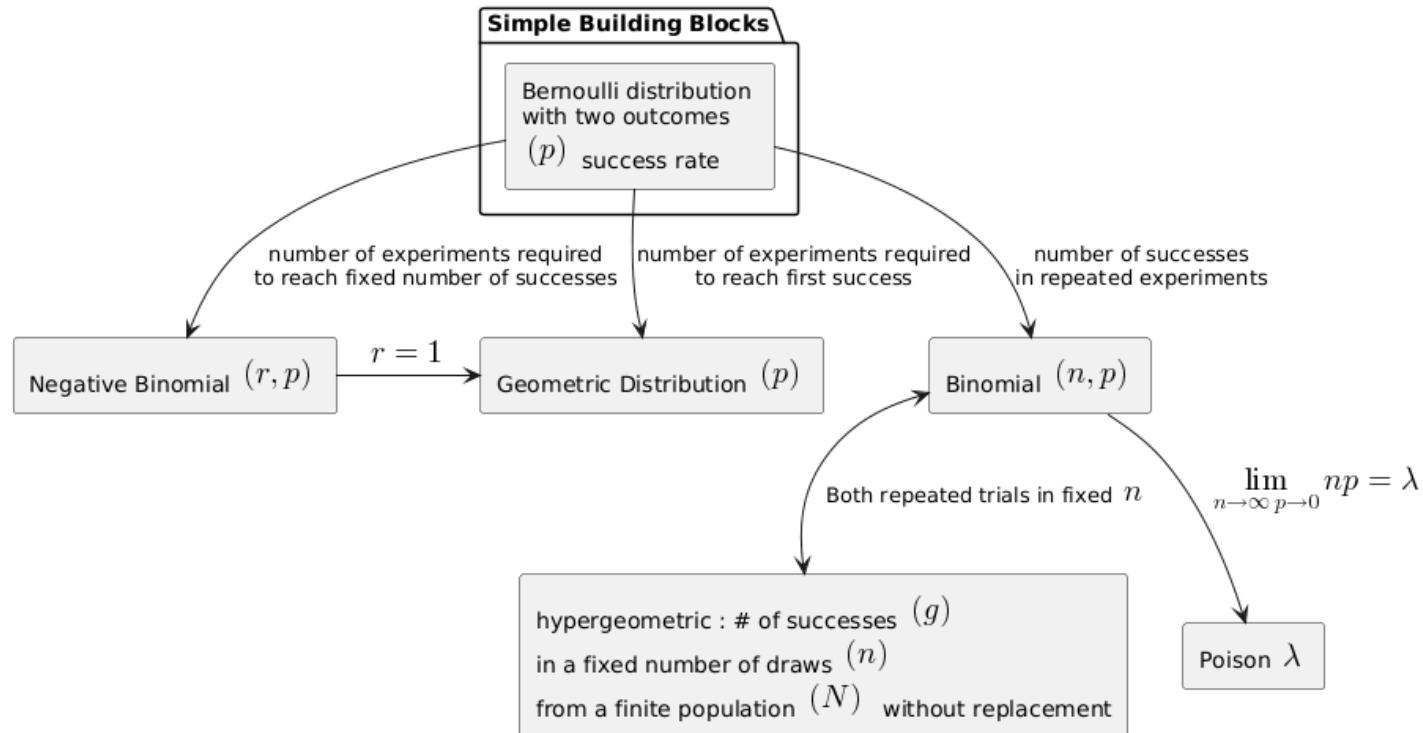
ii. 3. The Two Diagnostic Paths

A. Ord Plot: plots the relationship $k \frac{P(k)}{P(k-1)} (= ak + b)$. If the resulting plot is linear, the data belong to the Katz family (a subset of power series distribution). The specific distribution is identified by the signs of the slope (a) and intercept (b):

- Poisson: Slope is zero.
- Binomial: Slope is negative.
- Negative Binomial: Slope is positive and intercept is positive.
- Logarithmic Series: Slope is positive and intercept is negative.

B. Poissonness Plots: Plot a “count metameter” (which involves logs) against k . These are more robust because they use logs to isolate parameters like $\log(\lambda)$ as the slope.

1.3.8 "Family Tree" of Distributions



2

Joint Distribution

- Random variable in multiple dimensions (random vector)
- Model randomness in multiple dimensions
- The distribution of such “random vector”
- For 2 dimension random variable, the probability of (x, y) in a region is the volume of a 3 dimension object with the region as the base, under the “surface” defined by the joint density function $f(x, y)$

$$P((X, Y) \in A) = \int \int_A f(x, y) dy dx \quad \text{for all } A$$

$$P((X, Y) \in A) = \int \int_A f(x, y) dy dx$$

for all A

2.1 *Generalized Expectation of a Function of the Random Vector (with more than one dimension)*

$$E(g(X, Y)) = \int \int g(x, y) f(x, y) dx dy$$

The understanding of the generalized expected value of a function of random vector Note, for a special case of one dimension, the expectation of the random variable is just the expectation of the identity function. For 1 dimension, the aggregation of the identity function make sense, but it does not for more than one dimension.

2.2 Sample of problem solving with joint densities (exercises)

2.2.1 The probability of students being late to lecture

A Prob 140 student comes to lecture at a time that is uniformly distributed between 5:09 and 5:14. Independently of the student, the professor begins the lecture at a time that is uniformly distributed between 5:10 and 5:12. What is the chance that the lecture has already begun when the student arrives? –

To solve this problem using joint densities (Chapter 17), you use the property that if two random variables are independent, their joint density is the product of their marginal densities.

1. 1. Define the Marginals Let S be the student's arrival time and P be the professor's start time, measured in minutes after 5:00:

- Student: $S \sim \text{Uniform}(9, 14)$. The density is $f_S(s) = \frac{1}{14-9} = \frac{1}{5}$ for $s \in (9, 14)$.
- Professor: $P \sim \text{Uniform}(10, 12)$. The density is $f_P(p) = \frac{1}{12-10} = \frac{1}{2}$ for $p \in (10, 12)$.

2. 2. Identify the Joint Density Because S and P are independent, the joint density $f(s, p)$ is:

$$f(s, p) = f_S(s) \cdot f_P(p) = \frac{1}{5} \cdot \frac{1}{2} = \frac{1}{10}$$

This density is constant over a rectangle with a total area of $5 \times 2 = 10$.

3. 3. Calculate the Probability $P(P < S)$ Since the density is uniform, the probability is simply the Area of the region where $P < S$ (The students' arrival time S is after the arrival time of the professor P , assuming the professor starts lecture as soon as she arrives. "belated") divided by the Total Area of the possible outcomes.

- Region 1 (s between 10 and 12): In this 2×2 square, the line $p = s$ splits it in half. The area where $p < s$ is a triangle with area $\frac{1}{2}(2)(2) = 2$ (belated).
- Region 2 (s between 12 and 14): In this 2×2 rectangle, S is always greater than the maximum value of P (which is 12). The entire area is belated: $2 \times 2 = 4$.
- Total belated area: $2 + 4 = 6$. (See the picture in reMarkable)

4. Conclusion

$$\text{Chance} = \frac{\text{Belated Area}}{\text{Total Area}} = \frac{6}{10} = 0.6$$

There is a 60% chance the lecture has already begun when the student arrives.

Keys for joint distribution exercises:

1. Clarify on the region of the event of X, Y
2. The boundaries are defined by the equations (the limits) by replacing the inequality relationships by equality relationships from the inequalities defining the joint distribution's non-zero probabilities
3. Typically a triangle underneath a straight line
4. two typical points of $x = 0$ and $y = 0$ can determine the line
5. When performing integrals the bounds must obey the region boundaries.
6. for marginal density, also the boundaries apply in the integral and the applicable region of the marginal density

5. Alternative: solutions with integrals

To solve for the probability $P(P < S)$ using joint densities, we integrate the joint density function $f(s, p) = \frac{1}{10}$ over the region where $p < s$ within the possible bounds ($9 \leq s \leq 14$ and $10 \leq p \leq 12$) [17.1, 387, 389].

(a) 1. Formal Integration (Professor's time first) The most direct formulation is to integrate s with respect to p :

$$P(P < S) = \int_{10}^{12} \int_p^{14} \frac{1}{10} ds dp$$

(To reflect the constraint that $P < S$, so the minimum of the integral of ds is p)

(b) 2. Formal Integration (Student's time first) Because the region where $p < s$ is not a single simple rectangle relative to the student's arrival, the integral is split into two parts: Part 1: Student arrives while the professor might be starting ($10 \leq s \leq 12$). Part 2: Student arrives after the professor has definitely started ($12 \leq s \leq 14$). Note, the region where $S < P$ for ($9 \leq s \leq 10$) does not apply as that would be $S < P$ the opposite.

$$P(P < S) = \int_{10}^{12} \left(\int_{10}^s \frac{1}{10} dp \right) ds + \int_{12}^{14} \left(\int_{10}^{12} \frac{1}{10} dp \right) ds$$

(c) Calculus Breakdown Using the first expression:

i. Inner integral (ds): $\int_p^{14} \frac{1}{10} ds = \frac{14-p}{10}$

ii. Outer integral (dp): $\int_{10}^{12} \frac{14-p}{10} dp = \left[\frac{14p}{10} - \frac{p^2}{20} \right]_{10}^{12}$

iii. Evaluation:

$$\begin{aligned} & \left(\frac{14(12)}{10} - \frac{12 \cdot 12}{20} \right) - \left(\frac{14(10)}{10} - \frac{10 \cdot 10}{20} \right) \\ &= \frac{14(12) - 14(10)}{10} + \frac{-12 \cdot 12 + 10 \cdot 10}{20} \\ &= \frac{2 \cdot 14 \cdot 2 - 12 \cdot 12 + 10 \cdot 10}{20} \\ &= \frac{14 \cdot 2 - 6 \cdot 12 + 5 \cdot 10}{10} \\ &= \frac{2(14 - 3 \cdot 12 + 5 \cdot 5)}{10} \\ &= \frac{2(14 - 36 + 25)}{10} \\ &= \frac{2(3)}{10} = \frac{6}{10} = 0.6 \end{aligned}$$

The result is 0.6, matching our geometric solution.

2.2.2 JD: 2. For some constant c let X and Y have a joint density given by

$$f(x, y) = \begin{cases} c(x - y), & 0 < y < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

1. (a) Draw the region over which f is positive.

Your understanding is exactly correct.

The region where $f(x, y)$ is positive is indeed the **lower triangle** within the unit square [17.1]. To visualize it clearly, you can think of the triangle defined by these three vertices:

- $(0, 0)$
- $(1, 0)$
- $(1, 1)$

This area is bounded by the x-axis ($y = 0$), the vertical line $x = 1$, and the diagonal line $y = x$ [17.1].

2. (b) Find c .

(a) Solution summary:

- i. **Inner Integral (fixed x):** $\int_0^x c(x - y) dy = c \frac{x^2}{2}$.
- ii. **Outer Integral:** $\int_0^1 \frac{cx^2}{2} dx = \frac{c}{6}$.
- iii. **Normalization:** Setting $\frac{c}{6} = 1$ correctly yields $c = 6$.

(b) Detailed solution: Expressing the constraint $0 < y < x < 1$ as the boundaries of integral, with x first:

$$P((x, y) : 0 < y < x < 1) = \int_0^1 \int_0^x c(x - y) dy dx$$

Derivation assuming x constant:

$$\begin{aligned} & \int_0^x c(x - y) dy \\ & \left[c(xy - \frac{y^2}{2}) \right]_0^x \end{aligned}$$

$$\left[c\left(x - \frac{x^2}{2}\right) \right] - \left[c\left(0 - \frac{0^2}{2}\right) \right]$$

$$c \frac{x^2}{2}$$

then

$$\int_0^1 \left[c \frac{x^2}{2} \right] dx$$

$$\frac{c}{2} \left[\frac{x^3}{3} \right]_0^1$$

$$\frac{c}{6} 1 = \frac{c}{6}$$

therefore

$$\frac{c}{6} = 1$$

thus $c = 6$

3. (c) Find $P(X > Y + 0.4)$. Before you calculate, shade the event on the diagram you drew in (a).

Here is the reasoning to find the events of $X > Y + 0.4$ within the region of $0 < Y < X < 1$. The new boundary line is $X = Y + 0.4$, that is $X - 0.4 = Y$. Let $X' = X - 0.4$ and $Y' = Y + 0$, consider the coordinate translation to (X', Y') , in which $X' > Y'$. The origin of the new coordinates (X', Y') is at the $(0.4, 0)$ of the old coordinates (X, Y) , then the constraint of $X > Y + 0.4$ becomes $X' > Y'$. With similar reasoning as in (a). The region is to the right of the line of $X' = Y'$, a triangle with vertices of $(0.4, 0)$, $(1, 0)$, and $(1, 0.6)$ (for the last point, with $X' = Y' = Y$, $X' - X + 0.4 = 1.0 - 0.4 = 0.6$ for $X = 1.0$, thus the Y is 0.6).

Once the region (i.e. the boundaries of the events), using the same logic but different boundaries as in (b)

$$P(X > Y + 0.4) = \int_{0.4}^1 \int_0^{x-0.4} c(x-y) dy dx$$

Evaluations:

$$\int_0^{x-0.4} c(x-y) dy$$

$$\left[c\left(xy - \frac{y^2}{2}\right) \right]_0^{x-0.4}$$

$$\left[c\left(x(x-0.4) - \frac{(x-0.4)^2}{2}\right) \right]$$

Substituting $c = 6$,

$$3(x^2 - 0.16) = 3x^2 - 0.48$$

Integral with respect to dx :

$$\int_{0.4}^1 (3x^2 - 0.48) dx$$

$$[x^3 - 0.48x]_{0.4}^1$$

$$[x(x^2 - 0.48)]_{0.4}^1$$

$$[1(1^2 - 0.48)] - [0.4(0.4^2 - 0.48)]$$

$$0.52 + 0.128 = 0.648$$

4. (d) Find the density of X .

Is the marginal density $f_X(x)$ of X in the event space of A :

$$\int_{(x,y) \in E} f(x,y) dy$$

For the specific region where $0 < y < x < 1$, the marginal density $f_X(x)$ is defined as:

$$f_X(x) = \int_0^x f(x,y) dy, \quad 0 < x < 1$$

Evaluate and simplify:

$$\int_0^x 6(x-y) dy$$

$$\left[6xy - 6\frac{y^2}{2} \right]_0^x = [6xy - 3y^2]_0^x = 6xx - 3x^2 = 3x^2$$

5. (e) Are X and Y independent?

No, X and Y independent are not independent, as the values of Y is constrained by the values of X by $0 < y < x < 1$.

Because the support of the joint density is a triangle ($0 < y < x < 1$) rather than a rectangle, the possible values for Y are restricted by the specific value of X [17.2]. For X and Y to be independent, the joint density must be the product of the marginal densities ($f(x, y) = f_X(x)f_Y(y)$), and the support must be a rectangle with sides parallel to the axes [17.2].

Besides, by the marginal density function of $f_Y(y)$

$$f_Y(y) = \int_y^1 6(x-y) dx = [3x^2 - 6yx]_{x=y}^1 = (3 - 6y) - (3y^2 - 6yy) = 3y^2 - 6y + 3 = 3(y^2 - 2y + 1) = 3(y-1)^2$$

$$f_X(x)f_Y(y) = [3x^2][3(y-1)^2] = [3x(y-1)]^2 \neq 6(x-y)$$

where $f(x, y) = 6(x-y)$

6. (f) Find $E(XY)$. Your derivation is **entirely correct**.

Calculation Recap:

(a) Integrand: $6xy(x-y) = 6x^2y - 6xy^2$.

(b) Inner Integral (fixed x): $\int_0^x (6x^2y - 6xy^2) dy = [3x^2y^2 - 2xy^3]_0^x = x^4$.

(c) Outer Integral: $\int_0^1 x^4 dx = \left[\frac{x^5}{5} \right]_0^1 = \frac{1}{5}$.

Detailed derivations

$g(x, y) = xy$

$$E(XY) = \int_0^1 \int_0^x 6xy(x-y) dy dx$$

$$g(x, y)f(x, y) = 6x^2y - 6xy^2 = 6x(xy - y^2)$$

x being constant with integral with respect to y

$$6x \int_0^x (xy - y^2) dy = 6x \left[x \frac{1}{2} y^2 - \frac{1}{3} y^3 \right]_{y=0}^x = 6x \left[\frac{x^3}{2} - \frac{x^3}{3} \right] = x^4$$

$$E(XY) = \int_0^1 x^4 dx = \left[\frac{x^5}{5} \right]_{x=0}^1 = \frac{1}{5}$$

2.2.3 JD: 3. Let X and Y have joint density f given by

$$f(x, y) = \begin{cases} \frac{40}{243} x(3-x)(x-y), & 0 < y < x < 3 \\ 0 & \text{otherwise.} \end{cases}$$

Write the each of the following in terms of f but do not simplify the expression. Integrals should not include regions where $f(x, y) = 0$.

1. (a) $P(Y > 1)$

$$P(Y > 1) = \int_{x=1}^3 \int_{y=1}^x f(x, y) dy dx$$

since $0 < y < x < 3$ and $Y > 1$ thus we must have $1 < x < 3$ and $1 < y < x$ for each given x (The region of the event is the intersection of the original support ($0 < y < x < 3$) and the new constraint ($y > 1$). This defines a smaller triangle with vertices at $(1, 1)$, $(3, 1)$, and $(3, 3)$ [17.1].)

(a) (b) the conditional density of X given $Y = 1$ (please be clear about the values on which the density is positive)

Refresh:

$$P(A|B) = \frac{P(AB)}{P(B)}$$

The same way for the density function:

$$f_{X|Y=1}(x) = \frac{f(x, 1)}{f_Y(y = 1)}$$

$$f_Y(y) = \int_{x=y}^3 f(x, y) dx$$

The following is from Notebooklm

$$f_Y(y = 1) = \left[\int_{x=1}^3 f(x, 1) dx \right]$$

for $y = 1$

I'm not sure of the above. Why it is not the following: (First compute the marginal of y , then apply $y = 1$)

$$f_Y(y = 1) = \left[\int_{x=y}^3 f(x, y) dx \right] (y = 1)$$

maybe, they are the same?

therefore

$$f_{X|Y=1}(x) = \frac{f(x, 1)}{\left[\int_{x=1}^3 f(x, 1) dx \right]}$$

2. (c) $E(e^{XY}) g(x, y) = e^{XY}$

$$E(e^{XY}) = \int_{x=0}^3 \int_{y=0}^x e^{XY} f(x, y) dy dx$$

2.2.4 JD: 4. The joint density of X and Y is

$$f(x, y) = \begin{cases} 24xy, & x, y > 0 \text{ and } 0 < x + y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

1. (a) Find the density of X . Recognize this as one of the famous ones and state its name and parameters.

(a) The support for the joint density is a **triangle** in the first quadrant bounded by the axes ($x = 0, y = 0$) and the line $y = 1 - x$ [17.1].

Just a small correction to your coordinates:

- For $x = 0, y = 1$, so point A is $(0, 1)$.
- For $y = 0, x = 1$, so point B is $(1, 0)$.

The vertices of your region are $(0, 0)$, $(1, 0)$, and $(0, 1)$ [17.1].

More detailed thought process: First need to understand the region of the constraints for the non-zero values.

- i. $x, y > 0$, the region is in the first quadrant
- ii. $0 < x + y < 1$ The only meaningful boundary is defined by $x + y = 1$, as the other half of the condition $x + y = 0$ degenerates to both $x = 0$ and $y = 0$. The boundary is a line, as it's a linear equation. Given a line, finding two points on the line will tell how the line looks. Rearrange $y = -x + 1$ for the form of $ax + b$, the line's slope a is -1 , and for $x = 0 \Rightarrow y = 1$ thus point A $(0, 1)$ is on the line, and for $y = 0 \Rightarrow x = 1$, thus point B $(1, 0)$ is on the line. In the coordinates, then we can draw the line from A to B. With the visual, we can tell that the region is the triangle underneath and to the left of the line.

(b) Density of X Integrate over all possible Y subject to the constraints with respect to X :

$$f_X(x) = \int_{y=0}^{-x+1} 24xy \, dy = \left[24x \frac{y^2}{2} \right]_{y=0}^{-x+1} = 12x(1-x)^2$$

for $0 < x < 1$ this is Beta Distribution

- $r = 2, s = 3$

2. (b) Without further calculation, find the density of Y and justify your answer. From the above formula, swapping x to y :

$$f_Y(y) = \int_{x=0}^{-y+1} 24xy \, dx = \left[24y \frac{x^2}{2} \right]_{x=0}^{-y+1} = 12y(1-y)^2$$

for $0 < y < 1$ this is Beta Distribution

- $r = 2, s = 3$

because $f(x, y)$ and the constraints are all symmetry between x and y , thus the same logic applies to both x and y .

3. (c) Are X and Y independent? Why or why not?

No, they are not, since once the value of X is determined, the range of Y is constrained in the region at that particular point of X value. Vice versa for Y to X .

More details:

Because the support of the joint density is a triangle rather than a rectangle, the range of possible values for Y depends on the specific value of X [12 (17.2)]. Specifically, for any given x , Y is restricted to the interval $(0, 1 - x)$ [12 (17.1)]. Since the possible values of one variable change based on the other, they cannot be independent [12 (17.2)].

4. (d) Find the conditional density of X given $Y = 0.75$. As always, start with the possible values.

$$f_{X|Y=0.75}(x) = \frac{f(x, 0.75)}{f_Y}(0.75) = \frac{24x \cdot 0.75}{12 \cdot 0.75(1 - 0.75)^2} = 2 \cdot \frac{1}{0.25^2} x = 2 \cdot \frac{1}{\left(\frac{1}{4}\right)^2} x = 2 \cdot 16x = 32x$$

for $0 < x < 1 - 0.75$

The key justification, I'm using the above calculated $f_Y(y)$ with $y = 0.75$ substituting.

5. (e) Find $P(X > 0.2 | Y = 0.75)$. With the above calculated $f_X(X | Y = 0.75)$ and the applicable range, $P(X > 0.2 | Y = 0.75)$ is the integral in the range projected in the applicable range:

$$P(X > 0.2 | Y = 0.75) = \int_{0.2}^{0.25} 32x \, dx = 32 \left[\frac{x^2}{2} \right]_{0.2}^{0.25} = 16(0.25^2 - 0.2^2) = 16(0.25 - 0.2)(0.25 + 0.2) = 16 \cdot 0.05 \cdot 0.45 = 16 \cdot \frac{5}{100} \frac{45}{100} = 4 \frac{5}{25} \frac{9}{20} = 4 \frac{1 \cdot 9}{5 \cdot 20} = 1 \frac{9}{5 \cdot 5} = \frac{9}{25} = \frac{3}{10}$$

6. (f) Find $E(X | Y = 0.75)$.

Using the same scheme as in (e) with $g(x, y) = x$

$$E(X | Y = 0.75) = \int_0^{0.25} x \cdot 32x \, dx = 32 \left[\frac{x^3}{3} \right]_0^{0.25} = \frac{32}{3} \cdot 0.25^3 = \frac{32}{3 \cdot 4^3} = \frac{2}{3 \cdot 4} = \frac{1}{6} \approx 0.167$$

2.2.5 5. Let $U_i, 1 \leq i \leq 20$ be i.i.d. uniform $(0, 1)$ variables, and let $U_{(k)}$ be the k th order statistic.

- (a) What is the density of $U_{(7)}$?

(b) Without integrating the density, find the cdf of $U_{(7)}$.

[Draw a line representing the unit interval and put down crosses representing the variables. For $U_{(7)}$ to be less than x , how must you distribute the crosses?]

- (c) Find the joint density of $U_{(7)}$ and $U_{(12)}$.

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2.2.6 6. A random variable X has the beta $(2, 2)$ density. Given $X = x$, the conditional distribution of the random variable Y is uniform on the interval $(-x, x)$.

- (a) Find $P(Y < 0.2 | X = 0.6)$.

(b) Find $E(Y)$.

(c) Find the joint density of X and Y . Remember to specify the region where it is positive.

(d) Find $P(X < 0.3, |Y| < 0.3X)$. entered on: 2026-05-10 19:14:53

2.2.7 7. Let X and Y be i.i.d. with a joint density.

(a) Find $P(Y > X)$.

(b) Find $P(|Y| > |X|)$.

(c) If X and Y are i.i.d. standard normal, find $P(Y > |X|)$. entered on: 2026-05-10 19:14:56

2.2.8 8. Two points are placed independently and uniformly at random on the unit interval. This creates three segments of the interval.

What is the chance that the three segments can form a triangle?

[To find probabilities of events determined by two independent uniform $(0, 1)$ random variables, it's a good idea to draw the unit square.] entered on: 2026-05-10 19:15:16

3

Foundational Probability Calculations

3.1 Start from definitions

Perform reasoning and calculation definitions. This is often is the more straightforward approach that we don't use enough.

For example, the calculation of $E(X^2)$ where X is Bernoulli

$$E(X^2) = \sum_{\text{all possible values of } X^2} p_{X^2} X^2$$

3.2 Function application to random variable

$P(X = k) = P(f(X) = f(k))$ When you apply a 1-to-1 (injective) function f to a discrete random variable X , the probability of the new variable $Y = f(X)$ taking the value $f(k)$ is identical to the probability of X taking the value k .

3.2.1 Why this works

Because the mapping is 1-to-1, the event $\{X = k\}$ is logically equivalent to the event $\{f(X) = f(k)\}$. Since they represent the same set of outcomes in the sample space, they must have the same probability:

$$P(X = k) = P(f(X) = f(k))$$

3.2.2 Examples from our discussion

1. Squaring a Bernoulli: We previously saw that for a Bernoulli variable $X \in \{0, 1\}$, applying $f(X) = X^2$ is a 1-to-1 mapping on its domain. Therefore, $P(X^2 = 1) = P(X = 1) = p$ and $P(X^2 = 0) = P(X = 0) = 1 - p$

2. Monotone Transformations: In Data 140, we often use monotone functions (which are always 1-to-1) to transform variables. a function is not 1-to-1 ($f(X) = X^2$ for a variable that can be both positive and negative) and how that changes the probability? For example, if $X \in \{-1, 0, 1\}$, then $f(X) \in \{0, 1\}$ (3 elements in input domain to 2 elements in the outputs, both 1 and -1 mapped to 1, that is, the probability of $P(f(x) = 1) = P(X = -1) + P(X = 1)$)

3.3 Linearity of Expectations:

The expected value of a sum is the sum of the individual expectations, regardless of whether the variables are independent.

$$E \left[\sum_{i=1}^n \lambda_i X_i \right] = \sum_{i=1}^n \lambda_i E[X_i]$$

3.3.1 Applicaiton Example: Why is $Var(X) = E(X^2) - (E[X])^2$?

By definition: $Var(X) = E[(X - E[X])^2]$

By algebra expression (to figure out how to compute $(X - E[X])^2$ first, by $(a + b)^2 = (a - b)(a - b) = a^2 - 2ab - b^2$) $(X - E[X])^2 = X^2 - 2XE[X] + E[X]^2$

By linearity of expectation: $E[X^2 - 2XE[X] + E[X]^2] = E[X^2] - E[2XE[X]] + E[E[X]^2]$ For mean with respect to X , both $E[X]$ are constants, and $E[constant] = constant$ therefore $E[E[X]^2] = E[X]^2$ and $E[2XE[X]] = (2E[X])E[X] = 2E[X]^2$

By algeric substitution (replacing the equal value terms with simpler forms): $E[X^2 - 2XE[X] + E[X]^2] = E[X^2] - 2E[X]^2 + 2E[X]^2 = E[X^2] - E[X]^2$

3.3.2 Applicaiton Example: The mean of a Binomial(n, p) distribution is np because it is the sum of n Bernoulli(p) trials, each with mean p .

3.3.3 Variance and Covariance Rules: The variance of a sum includes the sum of individual variances plus twice the covariances of all pairs.

1. 1. Proof: Variance of a Sum The variance of a sum of random variables is derived from the definition of variance: $Var(X) = E[(X - E[X])^2]$

For a sum $S = \sum_{i=1}^n X_i$:

(a) Linearity of Expectation: $E[S] = \sum E[X_i]$.

- (b) Substitution: $Var(S) = E \left[\left(\sum (X_i - E[X_i]) \right)^2 \right]$ and let $(X_i - E[X_i]) = a_i$, then $Var(S) = E \left[\left(\sum a_i \right)^2 \right]$.
- (c) Expansion: Expanding the square of a sum $(\sum a_i)^2$ gives $\sum a_i^2 + 2 \sum_{i < j} a_i a_j$ (very elegant and convenient expression!), thus
- $$Var(S) = E \left[\sum a_i^2 + 2 \sum_{i < j} a_i a_j \right].$$
- (d) Expectation of Terms:
- The terms $E[(X_i - E[X_i])^2]$ are the individual variances, $\sum Var(X_i)$.
 - The cross-product terms $E[(X_i - E[X_i])(X_j - E[X_j])]$ are the covariances, $Cov(X_i, X_j)$.

Result: $Var(\sum X_i) = \sum Var(X_i) + 2 \sum_{i < j} Cov(X_i, X_j)$.

If the variables are independent, all covariance terms are zero, simplifying the formula to $Var(\sum X_i) = \sum Var(X_i)$.

3.4 MGF for Distribution Identification: The Moment Generating Function (MGF) of the sum of independent variables is the product of their individual MGFs.

Example: Multiplying the MGFs of two independent Poisson variables (μ and λ) results in the MGF for a Poisson($\mu + \lambda$).

3.5 Taylor Series Transformations: Using the expansion $e^z = \sum z^k / k!$ to simplify complex expectations into recognizable functional forms.

Example: This was used to simplify the expectation $E[e^{tX}]$ to derive the Poisson MGF $e^{\lambda(e^t - 1)}$.

4

Variance and Co-Variance

5

Optimization and Scaling

1. **Log-Transformations:** Converting products into sums to simplify differentiation, which is essential for Maximum Likelihood (MLE) and MAP estimates [History].
 - **Example:** Taking the natural log of $x^6(1-x)^4$ to find the mode of a Beta distribution via simpler addition-based derivatives [History].
2. **Derivative-Based Optimization:** Setting the first derivative of a density or log-density to zero to find the parameter value that maximizes the probability [History].
 - **Example:** Differentiating the log-posterior $\frac{6}{x} - \frac{4}{1-x} = 0$ to find the MAP estimate of 0.6 [History].
3. **Monotonic Function Application:** Since $\ln(x)$ is a monotonic increasing function, the value of x that maximizes $\ln(f(x))$ is guaranteed to be the same value that maximizes $f(x)$ [History].
 - **Example:** This allows us to ignore complex constants in the Beta density when finding the MAP [History].

6

Discrete and Bayesian Inference

1. **Recursive Odds Ratios:** Calculating $P(k)$ using the ratio $R(k) = \frac{P(k)}{P(k-1)}$, which avoids computationally heavy factorials for large n .
 - **Example:** To find $P(36)$ in a Binomial(60, 2/3), multiply $P(35)$ by the simple ratio $\frac{61-36}{36} \cdot 2$.
2. **Discrete Mode Identification:** Finding the peak of a histogram by identifying where the odds ratio $R(k)$ drops below 1.
 - **Example:** The mode of a Binomial(n, p) is the integer part of $(n + 1)p$.
3. **Conjugate Prior Updating:** Using the “Beta-Binomial” partnership to update beliefs by simply adding successes and failures to the prior parameters [22, History].
 - **Example:** A Beta(2, 3) prior combined with 2 heads in 7 tosses instantly becomes a Beta(4, 8) posterior [History].
4. **Order Statistics Mapping:** (Poisson Approximation)** **Method:** When n is very large and p is very small, you can approximate Binomial probabilities using the **Poisson** (λ) distribution, where $\lambda = np$. **Example:** Calculating the probability of a rare disease affecting exactly 5 people in a city of 100,000. Instead of a massive Binomial calculation, you use the Poisson PMF with $\lambda = 100,000 \cdot p$.

7

15. Poissonization

Method: A technique where you treat the total number of trials n as a random variable following a Poisson distribution. **Why it works:** This often “decouples” variables that are normally dependent. For example, if the number of people entering a store is Poisson, the number of men and women entering become **independent** Poisson variables.

16. Markov Chain Steady-State Analysis

Method: Modeling systems that transition between states where the next step depends only on the current state. You find the long-run “steady state” by solving the matrix equation $\pi P = \pi$. **Example:** Predicting the long-term market share of two competing brands based on monthly customer switching probabilities.

17. Central Limit Theorem (CLT) Approximation

Method: For a large sample size n , the distribution of the sample sum (or mean) is approximately **Normal**, regardless of the original distribution's shape. **Example:** Estimating the probability that the average height in a sample of 200 people is within a certain range using the Z -table.

10

18. Beta-Binomial Probability (The “C-Constant” Formula)

Method: Calculating the probability of k successes in n trials when the success probability p is unknown and follows a Beta(r, s) distribution. **Formula:** $P(S_n = k) = \binom{n}{k} \frac{C(r,s)}{C(r+k, s+n-k)}$ where C is the normalizing constant of the Beta density.

11

19. Multiple Linear Regression (Matrix Form)

Method: Finding the best-fit parameters β for a model with multiple predictors using the matrix identity $\hat{\beta} = (X^T X)^{-1} X^T Y$. **Example:** Predicting a house price (Y) based on square footage, age, and location (X_1, X_2, X_3) simultaneously.

12

Other pending topics to discuss

12.1 Covariance: $Cov(X_i, X_j) = E[(X_i - E[X_i])(X_j - E[X_j])]$

12.2 Unbiased Estimator:

When working with sample data, we use the divisor $(n - 1)$ in the formula $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ to ensure the estimate of σ^2 is unbiased.

12.2.1 2. Variance vs. Sample Variance

While the proof above describes the theoretical Variance (σ^2), statisticians often work with Sample Variance (S^2).

Feature	Variance (σ^2)	Sample Variance (S^2)
⋮	⋮	⋮
Nature	A fixed parameter of a population or distribution.	A statistic calculated from a finite sample of data.
Calculation	$E[(X - \mu)^2]$	$\frac{1}{n-1} \sum (X_i - \bar{X})^2$.
Goal	Describes the true underlying spread.	Estimates the true population variance.

3. Why use $(n - 1)$? (Unbiased Estimation) The sample variance S^2 is an unbiased estimator of σ^2 because its expected value equals the true population variance: $E[S^2] = \sigma^2$.

The “Bias” Problem: Using the population mean μ would allow a divisor of n . However, we use the sample mean $\bar{X}$, which is already “centered” in the data. This makes the deviations $(X_i - \bar{X})$ smaller on average than $(X_i - \mu)$. The Correction: Dividing by $n - 1$ (Bessel’s correction) perfectly compensates for this underestimation, ensuring the estimate is accurate on average.

12.3 Independent random variables

Two random variables X and Y are mathematically **independent** if the probability of any joint outcome is simply the product of their individual probabilities.

12.3.1 Mathematical Definitions

Discrete Variables: X and Y are independent if, for all possible values x and y :

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$$

. Continuous Variables: Independence occurs when the joint probability density function (PDF) is the product of the individual marginal densities:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

.

12.3.2 Key Properties in Data 140

1. **Expectation:** For independent variables, the expectation of their product equals the product of their expectations: $E[XY] = E[X]E[Y]$.
2. **Variance:** The variance of a sum of independent variables is simply the sum of their variances ($Var(X + Y) = Var(X) + Var(Y)$) because their **covariance is zero** [History].
3. **MGFs:** The Moment Generating Function of a sum of independent variables is the product of their individual MGFs: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$ [History].
4. **Conditional Probability:** If X and Y are independent, knowing the value of Y does not change the distribution of X : $P(X | Y) = P(X)$ [History].

In the **Bayesian framework** we discussed, we often move from assuming trials are “independent” to viewing them as **exchangeable**, where future trials depend on the evidence gathered from previous ones [History].

Would you like to see how this definition of independence is used to prove that the sum of independent **Normal** variables is also Normal?

Appendix: Hints of remembering the number of combinations formula $\binom{n}{k}$

Use the following equation:

$$n! = \binom{n}{k} k! (n - k)!$$

The total permualtions of n elements can be achieved by all possible k permutations and $(n - k)$ permutations.

$$n! = \binom{n}{k} k! (n - k)!$$

14

Core skills of mathematics required for probability

14.1 Calculus

1. Limit operation
2. Derivation
3. Integral
4. Function expression as series
 - (a) Taylor series
5. e

14.2 Linear algebra

14.3 Combinatory

14.4 Trigonometry

15

*If λ is not an integer, the mode of a Poisson distribution is the **floor** (the “bottom”) of λ , denoted as $\lfloor \lambda \rfloor$.*

Why? The reason is found in the **consecutive odds ratio**, which tells us the “direction” of the distribution’s shape:

$$R(k) = \frac{P(k)}{P(k-1)} = \frac{\lambda}{k}$$

Rising: As long as $k < \lambda$, the ratio $R(k) > 1$, meaning each probability is larger than the one before it. **Falling:** Once $k > \lambda$, the ratio $R(k) < 1$, and the probabilities begin to decrease.

If λ is not an integer, the highest point reached before the ratio drops below 1 is the largest integer less than λ (the floor). For example, if $\lambda = 2.7$, the probabilities rise until $k = 2$ and fall at $k = 3$, making **2** the mode.